HINT OF SOLUTION OF UNIT & DINMENSION

EXERCISE-1

PART - I

भाग - I

1.

	[1]P	A	S	C	[2]A	L							
	I				N					[3]W	A	[4]T	T
	C				G		[5]P			E		E	
	O				S		A			B		S	
	[6]M	H	O		[7]T	O	R	Q	U	E		L	
	E				R		S			R		A	
	[8]T	O	R	R	O		E		[9]M				
	R				[10]M	I	C	[11]R	O				
[12]F	E	R	M	I				A	L				
A					[13]C	A	N	D	E	L	A		
[14]R	P	M						I					
A								A					
[15]D	Y	N	E			[16]H	E	N	R	Y			

Answer is itself the solution. उत्तर ही स्वयं के लिए हल है।

2. We have the equation हमारे पास समीकरण है

$$\frac{Gm_1m_2}{r^2} = F$$

$$\frac{[G][M]^2}{[L]^2} = MLT^{-2}$$

$[G] = M^{-1}L^3T^{-2}$ ..... (i)

$\frac{hc}{\lambda}$ = Energy ऊर्जा

$\frac{[h]\ [c]}{[\lambda]} = ML^2T^{-2}$ $[c] = LT^{-1}$

$[\lambda] = L$

$[h] = ML^2T^{-1}$ ..... (ii)

$[c] = LT^{-1}$ ..... (iii)

taking the product of (i) & (ii) समीकरण (i) व (ii) का गुणन करने पर

$[G]\,[h] = L^5T^{-3}$

$[c]^3 = L^3T^{-3}$

$\therefore \quad \frac{[G][h]}{[c]^3} = L^2$

$G^{1/2}h^{1/2}c^{-3/2} = L$

again from (iii) दुबारा समीकरण (iii) से

$[T] = \frac{[L]}{[c]} = G^{1/2}h^{1/2}c^{-3/2-1} = G^{1/2}h^{1/2}c^{-5/2}$

From (ii) समीकरण (ii) से

$[h] = ML^2T^{-1}$

$[h] = \dfrac{MGhc^{-3}}{G^{1/2}h^{1/2}c^{-5/2}}$

$[h] = MG^{1/2}h^{1/2}c^{-3+5/2}$

or $G^{-1/2}\ h^{1/2}\ c^{1/2} = M$

3. All are dimensionally correct. विमीय रूप से सभी सत्य है।

PART - II

भाग - II

1. Velocity depends on length and time, so cannot be taken as base quantities.
वेग, दूरी और समय पर निर्भर करता है अतः इसे मूल राशि नही मान सकते है।

2. Angle is dimensionless but has unit (radian or degree)
कोण विमाहीन है लेकिन मात्रक रेडियन या डिग्री हो सकता है।

3. It is obvious. यह स्वतः स्पष्ट है।

4. [moment of force] = [F] [d] = ML^2T^{-2} .
[Moment of Inertia] = [I] = ML^2
[बलाघूर्ण] = [F] [d] = ML^2T^{-2} .
[जड़त्व आघूर्ण] = [I] = ML^2

5. $[v] = [k]\ [\lambda^a\ \rho^b\ g^c] \Rightarrow LT^{-1} = L^a\ M^b\ L^{-3b}\ L^c\ T^{-2c}$
$\Rightarrow LT^{-1} = M^b\ L^{a-3b+c}\ T^{-2c}$
$\Rightarrow a = ½, b = 0, c = ½$
so, $v^2 = kg\lambda$

6. $G = 6.67 \times 10^{-11}\ N\ m^2\ (kg)^{-2}$
$= 6.67 \times 10^{-11} \times 10^5$ dyne $\times 100^2\ cm^2 / (10^3)^2\ g^2 = 6.67 \times 10^{-8}$ dyne-cm^2-g^{-2}

7. It is obvious
यह स्वयं स्पष्ट है।

8. $[g] = LT^{-2}$ and numerical value $\propto \dfrac{1}{\text{unit}}$

$[g] = LT^{-2}$ और गणितीय मान $\propto \dfrac{1}{\text{मात्रक}}$

PART - III
भाग - III

1. $F = G\frac{m_1m_2}{r^2} \Rightarrow [G] = \frac{[F][r^2]}{[m_1m_2]} = \frac{MLT^{-2}L^2}{M^2}$

$= M^{-1} L^3 T^{-2}$

[Torque] [बलाघूर्ण] = [f] [d] = $MLT^{-2}L = ML^2T^{-2}$

[Momentum] [संवेग]= [m] [v] = MLT^{-1}

$[p] = \frac{[F]}{[A]} = \frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}$.

2. (i) $U = \sigma AT^4 \Rightarrow [\sigma] = \frac{[U]}{[A][T^4]} = \frac{ML^2T^{-3}}{L^2K^4} = MT^{-3} K^{-4}$

(ii) $\lambda T = b \Rightarrow [b] = [\lambda] [T] = LK$

(iii) $F = 6\pi\eta rv \Rightarrow [\eta] = \frac{[F]}{[r][v]} = \frac{MLT^{-2}}{L.\ LT^{-1}} = ML^{-1} T^{-1}$

(iv) $I = \frac{P}{A} = \frac{ML^2T^{-3}}{L^2} = ML^o T^{-3}$

(v) Energy ऊर्जा $= \frac{1}{2} Mi^2 \Rightarrow [M] = \frac{[E]}{[i^2]} = ML^2T^{-2} A^{-2}$

(vi) $\frac{[U]}{[V]} = \frac{[B^2]}{[2\mu_0]}$

$= [\mu_0] = \frac{[B^2][V]}{[U]}$

Also , $F = qVB \Rightarrow B = \frac{F}{qv}$

$[\mu_0] = \frac{(F)^2[V]}{[q^2v^2][U]} = MLT^{-2}A^{-2}$ **Ans.**

EXERCISE-2
PART - I

1. All the terms in the equation must have the dimension of force

$\therefore$ [A sin C t] = MLT^{-2}

$\Rightarrow$ $[A] [M^0L^0T^0] = MLT^{-2}$

$\Rightarrow$ $[A] = MLT^{-2}$

Similarly, $[B] = MLT^{-2}$

$\therefore \frac{[A]}{[B]} = M^0L^0T^0$

Again $[Ct] = M^0L^0T^0 \Rightarrow [C] = T^{-1}$

$[Dx] = M^0L^0T^0 \Rightarrow [D] = L^{-1}$

$\Rightarrow \frac{[C]}{[D]} = M^0L^1T^{-1}$.

समीकरण में सभी पदों की विमा बल की विमा होनी चाहिए

$\therefore$ $[A \sin C t] = MLT^{-2}$

$\Rightarrow$ $[A][M^0L^0T^0] = MLT^{-2}$

$\Rightarrow$ $[A] = MLT^{-2}$

इसी तरह, $[B] = MLT^{-2}$

$\therefore$ $\frac{[A]}{[B]} = M^0L^0T^0$

पुनः $[Ct] = M^0L^0T^0 \Rightarrow [C] = T^{-1}$

$[Dx] = M^0L^0T^0 \Rightarrow [D] = L^{-1}$

$\Rightarrow$ $\frac{[C]}{[D]} = M^0L^1T^{-1}$

2. $V = IR$

V has the dimensions of की विमा है

$$[V] = \frac{[\text{work}]}{[\text{charge}]} = \frac{ML^2T^{-2}}{AT} = ML^2\,T^{-3}\,A^{-1}$$

$$\therefore\ [R] = \frac{[v]}{[I]} = ML^2\,T^{-3}\,A^{-2}$$

$V = IR$

V has the dimensions of की विमा है

$$[V] = \frac{[\text{work}]}{[\text{charge}]} = \frac{ML^2T^{-2}}{AT} = ML^2\,T^{-3}\,A^{-1}$$

$$\therefore\ [R] = \frac{[v]}{[I]} = ML^2\,T^{-3}\,A^{-2}$$

3. $$\int \frac{x\,dx}{\sqrt{2ax - x^2}} = a^n \sin^{-1}\left[\frac{x}{a} - 1\right]$$

denominator $2ax - x^2$ must have the dimension of $[x]^2$

(∵we can add or substract only if quantities have same dimension)

हर $2ax - x^2$ की विमायें $[x]^2$ ही होनी चाहिये

(∵हम केवल तब ही जोड या घटा सकते हैं, जब राशियों की विमायें समान है।)

$$\therefore \left[\sqrt{2ax - x^2}\right] = [x]$$

Also, dx has the dimension of [x] dx की विमा भी वही है जो [x] की है।

$\therefore \frac{x\ \ dx}{\sqrt{2ax - x^2}}$ is having dimension L

$\therefore \frac{x\ \ dx}{\sqrt{2ax - x^2}}$ की विमा L है

Equating the dimension of L.H.S. & R.H.S. we have

दायें पथ व बायें पथ की विमायें समान करने पर

$[a^n] = M^0L^1T^0$ { $\therefore \sin^{-1}\left(\frac{x}{a} - 1\right)$ must be dimensionless विमाहीन होना चाहिये }

$\therefore n = 1$

4. $[\alpha] = \left[\frac{ma}{\beta}\right]$ ....(i)

$\left[\frac{\beta}{ma}\right][\ell] = M^0L^0T^0$

$\Rightarrow \quad \left[\frac{ma}{\beta}\right] = [\alpha] = [\ell] = L$

5. $[\alpha] = \left[\frac{e^2}{\varepsilon_0}\right]\left[\frac{1}{hc}\right]$

$= [Fr^2]\ \frac{1}{[E\lambda]}$

$= [M^1L^1T^{-2}L^2]\frac{1}{[M^1L^2T^{-2}L^1]} = [M^1L^3T^{-2}\ \ M^{-1}L^{-3}T^2] = [M^0\ L^0\ T^0]$

6. $\log y = m \log x + C$
$\log y = \log c' x^m$
$y = c'x^m$
$y = ax^b$

7. $Li = \frac{Li^2}{i} = \frac{Vq}{i}$ = volt – second

PART - II

1. $[b] = [V] = L^3$

$[a] = [RTV] = \frac{[PV]}{[n]} \cdot [V] = \frac{ML^2T^{-2}L^3}{mol}$

$= ML^5\ T^{-2}\ mol^{-1}$.

2. Let, $\omega = KM^ar^bG^c$ where K is a dimensionless constant.
Writing the dimension of both the sides and equating then we have,
$T^{-1} = M^aL^b(M^{-1}L^3T^{-2})^c$
$= M^{a-c}\ L^{b+3c}\ T^{-2c}$
Equating the exponents
$-2c = -1$ or $c = \frac{1}{2}$,
$b + 3c = 0$ or $-3c = b = -3/2$
$a - c = 0 \ .\ c = a = +1/2$

Thus the required equation is $\omega = K\sqrt{\frac{Gm}{r^3}}$

माना, $\omega = KM^ar^bG^c$ यहाँ K एक विमाहीन नियतांक है
दोनों ओर विमायें लिखते हुए बराबर करने पर
$T^{-1} = M^aL^b(M^{-1}L^3T^{-2})^c$
$= M^{a-c}\ L^{b+3c}\ T^{-2c}$
चरघातांकों को बराबर करने पर
$-2c = -1$ or $c = 1/2$,
$b + 3c = 0$ or $-3c = b = -3/2$
$a - c = 0 \ .\ c = a = +1/2$

पूछी गयी समीकरण है $\omega = K\sqrt{\frac{Gm}{r^3}}$

PART - III

1. All A, B & C are obvious.
A, B व C सभी स्वतः स्पष्ट है।

2. It is obvious

3. $[\alpha] = \frac{[h]}{[\sigma\theta^4]} = \frac{ML^2T^{-1}}{MT^{-3}K^{-4}.K^4} = L^2T^2$

So, unit of α will be m^2s^2.
अतः, α का मात्रक m^2s^2 होगा।

$$\frac{(\text{weber})\ (\Omega)^2(\text{Farad})^2}{\text{Tesla}} = \frac{Tm^2.\ \Omega^2F^2}{T} = m^2s^2$$

PART - IV

1. $[P] = \left[\frac{a}{V^2}\right] \Rightarrow [a] = [P]\,[V^2]$

2. $[b] = [V]$

3. $[a]\,[b] = [PV^2]\,[V] = [P]\,[V^3] = ML^{-1}T^{-2}\,[L^3]^3 = ML^8T^{-2}$

EXERCISE-3
PART - I
भाग - I

1. (A) $\frac{GM_eM_s}{R_e^{\,2}} = \text{Force}$

$[GM_eM_s] = [\text{Force}]\ [\text{बल}]\ [R_e^2]$
$= MLT^{-2}\,L^2 = ML^3T^{-2}$
Hence SI unit of GM_eM_s, will be (kilogram) (meter^3)(sec^{-2})
ie same as (volt) (coulomb) (metre)
अतः GM_eM_s का SI मात्रक होगा (किग्रा) (मी^{3}) (सेकण्ड$^{-2}$)
जो कि समान है (वोल्ट) (कूलाम) (मीटर)

(B) $\sqrt{\frac{3RT}{M}} = V_{R.M.S.}$

$\left[\frac{3RT}{M_0}\right] = [V_{R.M.S.}]^2 = L^2T^{-2}$

Hence SI unit will be (metre)^2 (second)$^{-2}$ ie same as (farad) (volt)^2 (kg)$^{-1}$
अतः SI मात्रक होगा (मी)^2 (सेकण्ड)$^{-2}$ जो कि समान है (फैरड) (वोल्ट)^2 (kg)$^{-1}$

(C) $\frac{[F^2]}{[q^2B^2]} = \frac{[q^2v^2B^2]}{[q^2B^2]} = [V^2] = L^2T^{-2}$

Hence SI unit (metre)^2 (second)$^{-2}$ i.e. same as (farad) (volt)^2 (kg)$^{-1}$
अतः SI मात्रक (मी)^2 (सेकण्ड)$^{-2}$ जो कि समान है (फैरड) (वोल्ट)^2 (kg)$^{-1}$

(D) $\left[\frac{GM_e}{R_e}\right] = \frac{[\text{Force}]\ [R_e]}{[\text{Mass}]} = \frac{MLT^{-2}L}{M} = L^2T^{-2}$

Hence SI unit will be (meter)$^{-2}$ (second)$^{-2}$ i.e. same as (farad) (volt)^2 (kg)$^{-1}$
अतः SI मात्रक (मी)^2 (सेकण्ड)$^{-2}$ जो कि समान है (फैरड) (वोल्ट)^2 (kg)$^{-1}$

2. (p) $U = \frac{1}{2}kT \Rightarrow ML^2T^{-2} = [k]\,K \Rightarrow [K] = ML^2T^{-2}K^{-1}$

(q) $F = \eta A\frac{dv}{dx} \Rightarrow [\eta] = \frac{MLT^{-2}}{L^2LT^{-1}L^{-1}} = ML^{-1}T^{-1}$

(r) $E = h\nu \Rightarrow ML^2T^2 = [h]\,T^{-1} \Rightarrow [h] = ML^2T^{-1}$

(s) $\frac{dQ}{dt} = \frac{kA\Delta\theta}{\ell} \Rightarrow [k] = \frac{ML^2T^{-3}L}{L^2K} = MLT^{-3}K^{-1}$

3. $d = k\,(\rho)^a\,(S)^b\,(f)^c$

$\Rightarrow [L] = \left[\frac{M}{L^3}\right]^a\left[\frac{M^1L^2T^{-2}}{L^2\ \ T}\right]^b\left[\frac{1}{T}\right]^c$

$0 = a + b$

$1 = -3a \Rightarrow a = -\frac{1}{3}$ So अतः $b = \frac{1}{3}$

$0 = -3b + c$

So अतः $n = 3$

4.* $M = h^x\,c^y\,G^z$

$M = (ML^2T^{-1})^x\,(LT^{-1})^y\,(M^{-1}L^3T^{-2})^z$

$x - z = 1$

$2x + y + 3z = 0$

$-x - y - 2z = 0$

$x = \frac{1}{2}, \quad y = \frac{1}{2}, \quad z = \frac{-1}{2}$

$M \alpha \sqrt{h}\sqrt{c}\,\frac{1}{\sqrt{G}}$

For L

$x - z = 0$

$2x + y + 3z = 1$

$-x - y - 2z = 0$

$x = \frac{1}{2} \quad y = \frac{-3}{2} \quad z = \frac{1}{2}$

$L \alpha \sqrt{h}\,\frac{1}{C^{3/2}}\,\sqrt{G}$

5.* (A) Energy of inductor प्रेरकत्व की ऊर्जा **=** $\frac{1}{2}LI^2 = \frac{1}{2}\frac{M_0N^2A}{\ell}I^2$

Energy of capacitor संधारित्र की ऊर्जा **=** $\frac{1}{2}CV^2 = \frac{1}{2}\in_0\frac{A}{d}V^2$

$\mu_0\frac{A}{\ell}I^2$ & $\in_0\frac{A}{d}V^2$ have same dimension समान विमा है

So इसलिये μ_0I^2 & $\in_0 V^2$ have same dimension समान विमा है

(C) $Q = CV$

$\frac{Q}{t} = \frac{CV}{t}$

$I = \in_0\frac{A}{\ell}\frac{V}{t}$

$\frac{A}{\ell t}$ have unit of speed चाल का मात्रक है

So $I = \in_0 cV$

6. $m = \frac{4\pi R^3}{3} \times \rho$

$\ell n(m) = \ell n\left(\frac{4\pi}{3}\right) + \ell n(\rho) + 3\ell n(R)$

$0 = 0 + \frac{1}{\rho}\frac{d\rho}{dt} + \frac{3}{R}\frac{dR}{dt}$

$\Rightarrow \left(\frac{dR}{dt}\right) = v \propto -R \times \frac{1}{\rho}\left(\frac{d\rho}{dt}\right)$

$v \propto R$

7. In terms of dimension
विमा के पदों में

$qE = qvB$

$E = vB$

$[E] = [B]\,[LT^{-1}]$

8. $C = \frac{1}{\sqrt{\mu_0 \in_0}}$

$C^2 = \frac{1}{\mu_0 \in_0}$

$\mu_0 = \in_0 . C^2$

$[\mu_0] = [\in_0^{-1}]^{-1} L^{-2} T^2$

PART - II
भाग - II

1. Energy stored in inductor
प्रेरक मे संचित ऊर्जा

$U = \frac{1}{2}LI^2 \qquad \Rightarrow \qquad L = \frac{2U}{I^2}$

$[L] = \frac{ML^2T^{-2}}{Q^2/T^2} = \frac{ML^2}{Q^2}$

Since Henry is unit of inductance L
प्रेरक L का मात्रक हैनरी है।

$\therefore$ (4) is correct. सही है।

2. From F = qvB से

$\Rightarrow$ $[MLT^{-2}] = [C]\,[LT^{-1}]\,[B]$

$\Rightarrow$ $[B] = [MC^{-1}T^{-1}]$

3. $F = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{R^2}$

$\varepsilon_0 = \frac{q_1q_2}{4\pi FR^2}$

Hence $\varepsilon_0 = \frac{C^2}{N.m^2} = \frac{[AT]^2}{MLT^{-2}.L^2} = [M^{-1}L^{-3}T^4A^2]$

Ans. (2)

4. volume of man becomes = $(9)^3$ times
weight of man becomes = 9^3 times
Cross section area in leg = 9^2 times

$$\text{stress} = \frac{\text{weight}}{\text{Area}} = 9 \text{ times}$$

आदमी का आयतन = $(9)^3$ गुना हो जायेगा
आदमी का भार = 9^3 गुना हो जायेगा
पैर का अनुप्रस्थ काट क्षेत्र = 9^2 गुना हो जायेगा

$$\text{प्रतिबल} = \frac{\text{भार}}{\text{क्षेत्रफल}} = 9 \text{ गुना हो जायेगा}$$

